



On-the-Job Training Report

Data Engineer at LTSS

This is to certify that Mr. Pankaj Kumar Modi of Bhaskar Waman Thakur College of Science (VIVA) is employed as a Data Engineer at Luppiter Tech Solutions & Services while pursuing his M.Sc. in Computer Science under the University of Mumbai. The details of his employment are as follows:

Employment Start Date: 12/09/2025

Current Status: Currently Employed

Broad Area of Work: Data Engineering, ETL Development, Data Integration, and Process Automation

Brief Description of Work:

Mr. Pankaj Kumar Modi is working as a Data Engineer at **Luppiter Tech Solutions & Services**, where he is involved in designing, developing, and maintaining enterprise-grade ETL (Extract, Transform, Load) pipelines for large-scale data integration projects. His responsibilities include participating in ETL migration initiatives, automating repetitive and manual processes to improve operational efficiency, and ensuring reliable data movement across enterprise systems. He also collaborates with cross-functional teams, contributes to technical solution development, and leads team activities to ensure project objectives are achieved within the stipulated timelines while maintaining high standards of quality and performance. He is currently serving in this role as a full-time employee.

Signature:

Name: Jignesh Chheda

Designation: Senior Data Architect

Email: jignesh.c@luppiter.tech

Professional Evaluation

Name of Employee: Pankaj Kumar Modi

College/Institution: Bhaskar Waman Thakur College of Science (VIVA)

[Note: Give a score on a scale of 1 to 5 by putting ✓ in the respective cells.]

No.	Particulars	Excellent	Very Good	Good	Moderate	Satisfactory
1	Attendance & Punctuality	✓				
2	Ability to Work in a Team	✓				
3	Written and Oral Communication Skills		✓			
4	Problem-Solving Skills	✓				
5	Ability to Grasp New Concepts		✓			
6	Technical Skills (Data Engineering, ETL Development, Automation, Programming)	✓				
7	Ability to Complete Assigned Tasks	✓				
8	Quality of Overall Work Done	✓				

Comments:

Mr. Pankaj Kumar Modi has consistently demonstrated strong technical expertise, professionalism, and a commitment to excellence in his role as a Data Engineer at Luppiter Tech Solutions & Services. He has made valuable contributions to the design and development of enterprise ETL pipelines, ETL migration initiatives, and the automation of repetitive processes to improve operational efficiency.

Signature:

Name: Jignesh Chheda

Designation: Senior Data Architect

Data Engineer at Luppiter Tech Solutions & Services

Title:

Design, Development, and Automation of Enterprise Data Engineering Pipelines Using Azure Data Factory, Databricks, PySpark, and Delta Lake

Description of the Organization:

Luppiter Tech Solutions & Services (LTSS) is a technology consulting and digital transformation company specializing in enterprise data solutions, performance management consulting, ETL services, application development, staff augmentation, and technical training. The company helps organizations transform raw data into actionable business insights through innovative, data-driven solutions and modern cloud technologies.

With expertise across industries such as Banking & Financial Services (BFSI), Pharmaceuticals, Retail, Manufacturing, and Oil & Gas, LTSS delivers scalable, cost-effective, and high-performance technology solutions. Headquartered in Singapore, with delivery centers in Mumbai, India, and Auckland, New Zealand, the organization serves global clients by providing end-to-end consulting, cloud-based data engineering, process automation, and enterprise application development services.

Description of the Activities

The Data Engineering team is responsible for designing, developing, and optimizing cloud-based data pipelines that enable efficient data ingestion, transformation, and integration across enterprise systems. The team builds scalable ETL solutions using Azure Data Factory, Azure Databricks, PySpark, and Delta Lake while ensuring data quality, reliability, and performance through modern data engineering practices.

As a Data Engineer, my responsibilities include developing end-to-end ETL pipelines, implementing the Medallion (Bronze, Silver, Gold) architecture, performing data transformation and validation using PySpark, orchestrating workflows with Azure Data Factory, and optimizing pipeline performance. I collaborate with business analysts, solution architects, testers, and cross-functional teams to deliver scalable, reliable, and production-ready data solutions.

Description of Work Allotted

As a Data Engineer Trainee at Luppiter Tech Solutions & Services, I gained hands-on experience in modern data engineering practices and cloud-based ETL development. During my training, I worked with Azure Data Factory, Azure Databricks, PySpark, and Delta Lake to understand how enterprise data pipelines are developed, managed, and optimized. I participated in data integration activities, assisted in building ETL workflows, and learned industry-standard approaches for processing and managing large datasets. The major activities carried out during my training are as follows:

- Learned the fundamentals of enterprise ETL (Extract, Transform, Load) processes and the complete data integration lifecycle.
- Assisted in developing and maintaining ETL pipelines using Azure Data Factory and Azure Databricks.
- Performed data extraction, cleaning, transformation, validation, and loading using PySpark.
- Gained practical exposure to the Medallion Architecture (Bronze, Silver, and Gold) and implemented data processing using Delta Lake.
- Worked with Parquet and Delta file formats to understand efficient data storage and processing techniques.
- Learned Spark concepts such as DataFrames, transformations, actions, partitioning, and caching while processing large datasets.
- Assisted in monitoring ETL workflows, validating outputs, and troubleshooting pipeline failures to ensure successful execution.
- Participated in testing and debugging data pipelines to improve data quality and reliability.
- Used Git for version control, source code management, and collaboration with team members.
- Developed reusable PySpark notebooks and followed coding best practices to improve maintainability and readability.
- Gained knowledge of Data Lake, Data Warehouse, and cloud-based data engineering concepts used in enterprise environments.
- Explored the use of AI tools and Large Language Models (LLMs) to support code understanding, documentation, debugging, and productivity during development activities.
- Collaborated with senior Data Engineers and cross-functional teams to understand business requirements and implement scalable data engineering solutions.

WEEK 1

Day	Date	Name of Topic / Module Completed	Remarks
Monday	06/04/2026	Introduction to Data Engineering & Project Overview	Understood the project architecture, business requirements, ETL lifecycle, and Azure-based data engineering workflow.
Tuesday	07/04/2026	Azure Data Factory Fundamentals	Learned Azure Data Factory components, linked services, datasets, pipelines, and integration runtimes.
Wednesday	08/04/2026	Azure Databricks Environment Setup	Configured Databricks workspace, clusters, notebooks, and workspace structure for development.
Thursday	09/04/2026	Understanding Source Data	Explored source datasets, data formats (CSV, JSON), schema analysis, and business rules.
Friday	10/04/2026	Git and Project Repository	Learned Git workflow, repository structure, branching strategy, and version control practices.

Signature of faculty mentor: _____

Seal of the University

WEEK 2

Day	Date	Name of Topic / Module Completed	Remarks
Monday	13/04/2026	PySpark Basics	Learned DataFrames, SparkSession, transformations, actions, and basic PySpark operations.
Tuesday	14/04/2026	Data Cleaning using PySpark	Removed null values, handled duplicates, standardized data, and applied business validations.
Wednesday	15/04/2026	Data Transformation	Performed filtering, joins, aggregations, derived columns, and data enrichment.
Thursday	16/04/2026	Reading and Writing Data	Worked with CSV, Parquet, and Delta formats for efficient data storage.
Friday	17/04/2026	PySpark Practice	Developed reusable notebooks and validated transformed datasets.

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WEEK 3

Day	Date	Name of Topic / Module Completed	Remarks
Monday	20/04/2026	Azure Data Factory Pipeline Development	Created pipelines for ingesting source data into the Data Lake.
Tuesday	21/04/2026	Parameterization in ADF	Implemented dynamic pipelines using parameters and variables.
Wednesday	22/04/2026	Trigger and Scheduling	Configured scheduled triggers and monitored pipeline executions.
Thursday	23/04/2026	Error Handling	Implemented retry policies, logging, and failure handling in pipelines.
Friday	24/04/2026	Pipeline Testing	Validated successful data ingestion and workflow execution.

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WEEK 4

Day	Date	Name of Topic / Module Completed	Remarks
Monday	27/04/2026	Bronze Layer Development	Loaded raw source data into the Bronze layer using Delta Lake.
Tuesday	28/04/2026	Silver Layer Development	Cleaned, validated, and transformed data into standardized datasets.
Wednesday	29/04/2026	Gold Layer Development	Created business-ready aggregated datasets for reporting and analytics.
Thursday	30/04/2026	Delta Lake Concepts	Learned ACID transactions, versioning, and efficient data management.
Friday	01/05/2026	Medallion Architecture Review	Validated complete Bronze-Silver-Gold workflow.

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WEEK 5

Day	Date	Name of Topic / Module Completed	Remarks
Monday	04/05/2026	End-to-End ETL Development	Connected Azure Data Factory with Databricks notebooks for complete ETL execution.
Tuesday	05/05/2026	Workflow Orchestration	Managed dependencies and sequencing between ETL activities.
Wednesday	06/05/2026	Data Validation	Compared source and target records and verified data accuracy.
Thursday	07/05/2026	Performance Improvement	Learned partitioning, caching, and optimization techniques in PySpark.
Friday	08/05/2026	Pipeline Monitoring	Reviewed execution logs and resolved basic pipeline issues.

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WEEK 6

Day	Date	Name of Topic / Module Completed	Remarks
Monday	11/05/2026	Incremental Data Processing	Implemented incremental data loading using business keys and timestamps.
Tuesday	12/05/2026	Data Quality Checks	Applied validation rules to ensure consistency and completeness of data.
Wednesday	13/05/2026	Exception Handling	Implemented logging and handled invalid records during ETL execution.
Thursday	14/05/2026	Reusable Notebook Development	Created modular PySpark notebooks for reuse across pipelines.
Friday	15/05/2026	Project Integration Testing	Tested end-to-end workflow and validated pipeline execution.

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WEEK 7

Day	Date	Name of Topic / Module Completed	Remarks
Monday	18/05/2026	Data Warehouse Concepts	Understood fact tables, dimension tables, and star schema design.
Tuesday	19/05/2026	Reporting Dataset Preparation	Generated business-ready datasets for analytics and reporting.
Wednesday	20/05/2026	Git Collaboration	Managed code commits, pull requests, and version updates.
Thursday	21/05/2026	Documentation	Prepared technical documentation for ETL pipelines and workflows.
Friday	22/05/2026	Code Review	Participated in code review and incorporated feedback from senior engineers.

Signature of faculty mentor: _____

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WEEK 8

Day	Date	Name of Topic / Module Completed	Remarks
Monday	23/05/2026	End-to-End Project Validation	Validated complete ETL workflow from source ingestion to Gold layer generation.
Tuesday	24/05/2026	Pipeline Optimization	Fine-tuned pipeline execution and verified processing performance.
Wednesday	25/05/2026	Project Demonstration	Demonstrated the end-to-end Azure Data Engineering solution to the project team.
Thursday	26/05/2026	Documentation and Knowledge Transfer	Updated project documentation and summarized key implementation details.
Friday	27/05/2026	Final Review and Project Closure	Reviewed learning outcomes, resolved pending observations, and completed project activities.

Signature of faculty mentor: _____

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Self-Assessment:

Working as a Data Engineer Trainee at Luppiter Tech Solutions & Services gave me valuable practical exposure to modern data engineering technologies and cloud-based ETL development. During this training, I worked with Azure Data Factory, Azure Databricks, PySpark, Delta Lake, Python, SQL, and Git. I learned how enterprise ETL pipelines are developed, how data is extracted, transformed, validated, and loaded, and how cloud-based data engineering solutions are implemented. This experience enhanced my technical knowledge, problem-solving skills, and understanding of industry-standard development practices. It also improved my ability to work in a collaborative team environment and understand real-world project requirements.

Conclusion:

My training at Luppiter Tech Solutions & Services was a valuable learning experience that helped me connect my academic knowledge with practical industry applications. I gained hands-on experience in developing ETL pipelines using Azure Data Factory and Azure Databricks, performing data transformation and validation using PySpark, and working with Delta Lake and the Medallion Architecture. I also learned the importance of version control using Git, data quality, and efficient pipeline execution. Overall, this training strengthened my foundation in Data Engineering and increased my confidence to work on real-world data engineering projects and continue growing as a Data Engineer.

Signature:

Name: Jignesh Chheda

Designation: Senior Data Architect

Email: jignesh.c@luppiter.tech